Enforcement Patterns and Policy Shifts in U.S. Immigration: A Data-Driven Analysis

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GitHub: github.com/Jackie995/Advanced_Data_Science_Project **Video:** youtu.be/YRulkxlKT_s

Introduction and Background

For this project, we analyzed the possible causes and effects of immigration enforcement within the United States. As a group composed of immigrants, we have firsthand experience with the complexity of the US immigration system, which drove us to look beyond just the surface level headlines and dive deeper into the numerical trends behind enforcement and their impact on factors such as a migrator's aspirations to move.

We looked at data from U.S. Immigration and Customs Enforcement (ICE), the Deportation Data Project, the Department of Homeland Security (DHS), the US State Department, and the World Bank API to determine whether changes in enforcement data correlate with policy changes or external global factors.

Research Questions

We originally started this project with three research questions:
- What factors affect immigration the most?
- Do fewer people immigrate when removal (deportation) rates are high?
- Has the US immigration system become more enforcement-focused or admission-focused?

We also expanded our analysis to also look into the GDP per capita of origin countries, Visa issuance trends, and country-level removal patterns alongside the original questions with the cohesive goal of understanding what enforcement patterns correlate with - whether that be policy events, economic conditions, or admission trends.

Methods

Data Sources

We used five datasets in our project. The ICE, DHS, and World Bank API Net Migration datasets were imported from Kaggle, the Visa statistics were obtained directly from the US Department of State, and the GDP per capita data was pulled from the official World Bank API website, as Kaggle did not have that specific dataset.

The ICE dataset contains immigration enforcement activity across approximately ten years, with key features including Apprehension Date, Departure Date, Departure Country, and Case Status.

These features enabled us to use aggregation on the arrests, detentions, and removals datasets and map them to a yearly level. The DHS immigration statistics dataset spans from 1980 to 2021 and captures the annual counts of the Obtainment of Permanent Residence, Refugee Arrivals, Noncitizen Apprehensions, Removals, and Returns. The World Bank API net migration dataset provides migration estimates per country over time, with features such as Country Name, Year, and Net Migration count. The US department of State Visa statistics contains annual Visa issuance counts by country, which we aggregate across all issuing offices to produce a single value per country annually. Finally, the World Bank GDP per capita dataset records the economic output per person by country and year- which we used to explore relationships between economic conditions in departure countries and ICE removal counts.

Data Cleaning and Preparation

The three datasets imported from Kaggle required minimal cleaning aside from selecting relevant features and date ranges for our analysis. The ICE data was filtered to 2012 since the dataset starts from late 2011, meaning that the incomplete data would lead to distortions if included. The merged DHS immigration dataset was filtered to 2012 to 2021 to align with the ICE data window, and all numeric columns stored were converted to mathematically usable types such as floats or integers. For the Visa statistics datasets, records from multiple issuing offices within each country were summed into a single annual value per country.

Methods of Analysis

We used two primary data science methods: Principal Component Analysis (PCA), and K-Means clustering.

PCA was applied to the DHS immigration dataset to reduce dimensionality of our dataset and reveal its underlying patterns among the five immigration features (Obtainment of Permanent Residency, Refugee Arrivals, Noncitizen Apprehensions, Removals, and Returns). All features were then standardized using a Standard Scaler before PCA was applied. We ran PCA on three time windows: the full 1980 - 2021 period (excluding outlier years such as COVID), the Obama-era window of 2009 - 2017, and the Trump-era window of 2017 - 2021. For each window we analysed the PC1 and PC2 loadings to understand which features drive each component, and we plotted each feature's contribution to PC1 over time to visualize year-specific shifts within the dataset. We then extended the PCA into a Principal Component Regression (PCR) model to predict annual permanent presidency obtainment from principal components for the general time period 1980 - 2021.

K-Means clustering was applied to the monthly ICE enforcement data (Removals, Arrests, and Detentions) in an attempt to identify specific distinct enforcement behavior regimes over time. Features were standardized before clustering, and we used the elbow method to determine that k=3 was the optimal number of clusters, assigning interpretive labels to each cluster based on their centroid values: low-intensity, moderate-intensity, and high-intensity enforcement periods.

Results

K-Means Clustering of ICE Data

To determine the number of clusters, we applied the elbow method from k = 1 through k = 9. The plot below shows that a clear inflection point is at k = 3, where the rate of inertia visibly flattens.

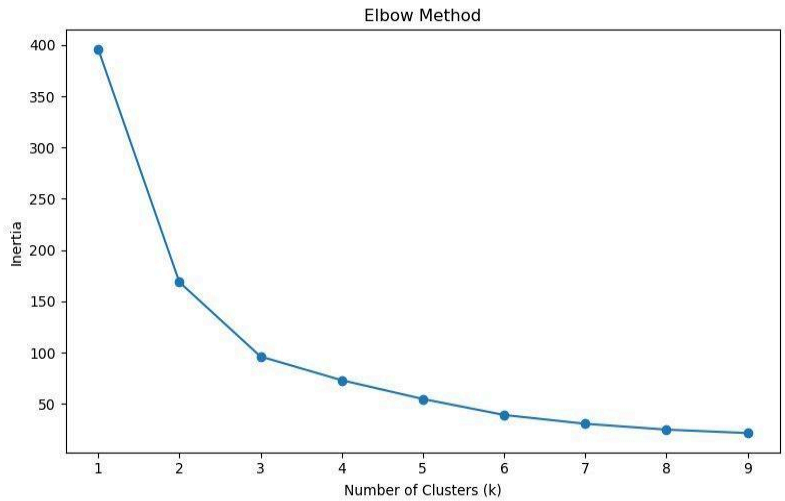

Figure 1. Elbow Method for K-Means Cluster Selection. Inertia flattens at k=3, indicating three distinct enforcement regimes in the monthly ICE data.

With k = 3, the cluster scatterplot of removals versus detentions reveals three well separated enforcement period groups. The low intensity yellow cluster is in the bottom left, representing months with lower removals and detention counts. The moderate intensity purple cluster takes over the central part of the graph, and the high intensity teal cluster extends towards the upper right, representing months with the highest combined enforcement activities. This separation between these clusters is the start to answering our first question as rather than looking at individual metrics, the clusters let us see enforcement from a holistic view that either ramps up or pulls back as a whole.

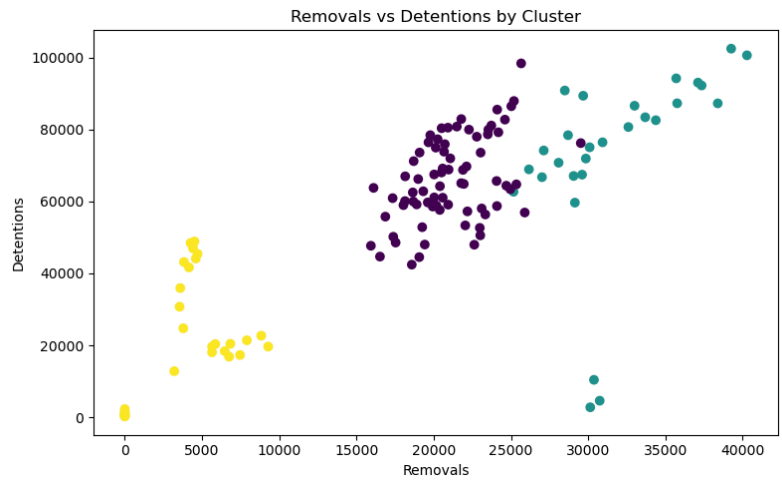

Figure 2. ICE Removals vs. Detentions by Cluster. Three clusters correspond to distinct enforcement intensity levels: low (yellow), moderate (purple), and high (teal).

The arrests vs. removals scatter plot tells a complementary story. In the high-intensity teal cluster, arrests and removals scale tightly together, meaning nearly every arrest is being converted into a formal removal order. In the low-intensity yellow cluster, arrests are low and the few removals that do occur are not strongly correlated with arrests suggesting that during low-enforcement periods, removals are being driven by factors other than new arrests (such as the processing of older cases). This distinction matters for understanding whether the system is being aggressive or simply clearing backlogs.

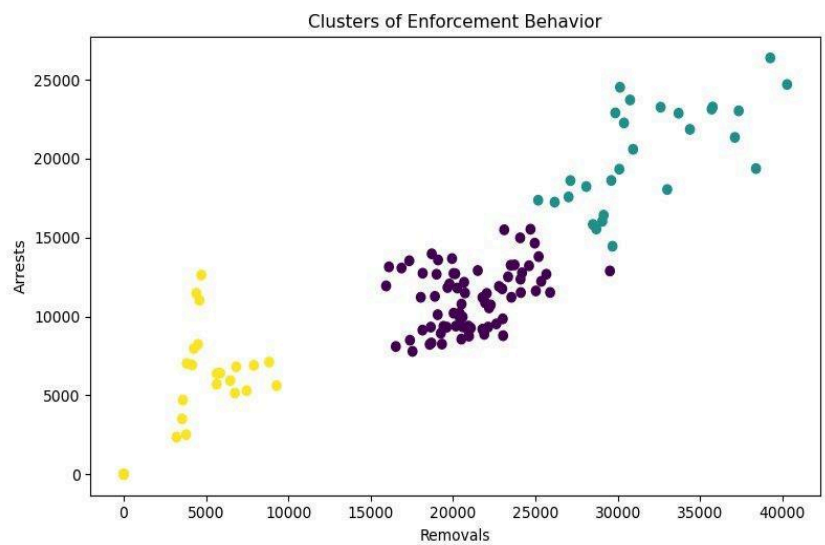

Figure 3. Clusters of Enforcement Behavior: ICE Removals vs. Arrests. The three enforcement intensity tiers are clearly separated, with the high-intensity cluster (teal) showing the tightest arrest-to-removal conversion rate.

The cluster assignment timeline is where the enforcement story connects directly to policy. Rather than asking whether enforcement went up or down in a given year, we can now ask: did the system fundamentally change its operating mode, and if so, when?

Figure 4. ICE Cluster Assignment Over Time. Cluster assignments shift from high-intensity (2012) to moderate-intensity (2012–2014), to low-intensity (2014-2020) consistent with COVID-19 disruptions in 2020.

Three patterns stand out. First, cluster 2 which is a high intensity appears only in early 2012 and then again from 2020 and onward. The 2012 peak aligns with two overlapping policy drivers: the Secure Communities Program (Noferi, 2014) and the Confidence Delivery System that went into effect in 2012 (Capps et al., 2017). The Secure Communities Program linked local law enforcement fingerprint databases with federal immigration records, enabling minor arrests to trigger immigration checks and potential deportations. The CDS went into effect in 2012 and shifted the government away from quick voluntary return towards formal removals, directly inflating removal counts. This answers our first question, in 2012 it was policy infrastructure, not simply political will, that drove enforcement to its peak.

The second thing is cluster 1, with a moderate intensity, covers roughly 2012-2014, a transition period where removals remain relatively high but detention begins to moderately increase. DACA's implementation in 2012 (U.S. Citizenship and Immigration Services, n.d.) gave undocumented individuals who came to the country as children temporary protection from removal. Enforcement did not collapse immediately, but it began losing its earlier broad intensity as priorities started to shift. This pattern directly speaks to whether fewer people face removals when a protective policy is in place.

The third and final pattern is that the low intensity cluster 0 dominates from 2014 to 2020. After the 2013 peak, the Obama administration narrowed enforcement priorities to individuals with serious criminal records, and overall removal numbers fell noticeable. The hard return to cluster 0 in 2020 is not a policy shift but a disruption with COVID-19. The fact that both a deliberate policy change in 2014 and an external catastrophe in 2020 produced the same low-enforcement cluster assignment is telling that the system looks the same from the outside regardless of why it slowed down.

ICE Enforcement Activity Over Time

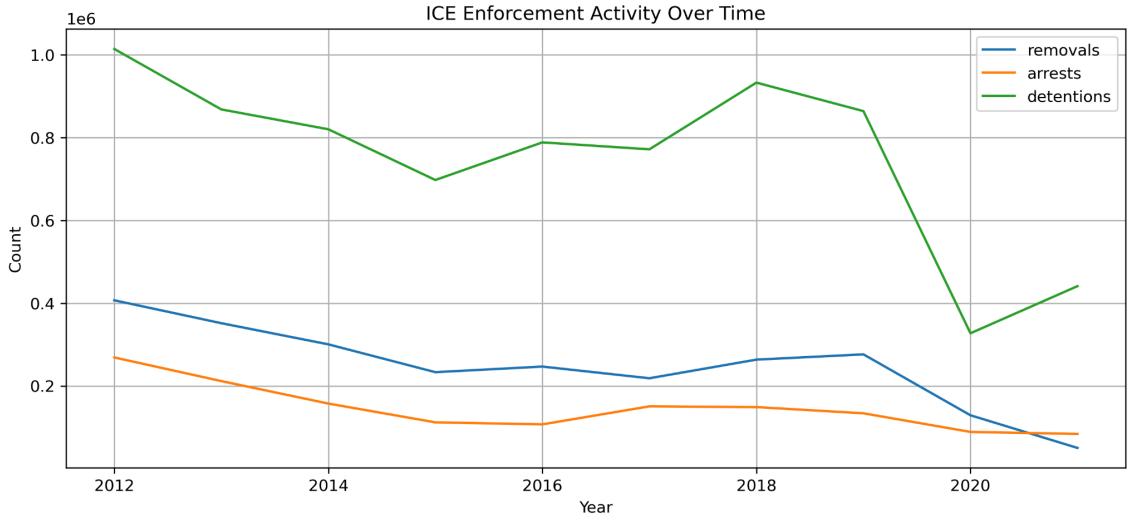

Figure 5. ICE Enforcement Activity Over Time (Annual). Detentions (green) are consistently the highest-volume metric. Arrests (orange) and removals (blue) follow a gradual decline through 2019 before dropping sharply in 2020.

This plot extends the clustering picture by showing each enforcement metric separately on an annual basis. Detentions are consistently the highest-volume metric, which makes structural sense: individuals who are arrested are often detained before a removal decision is made, so detentions will naturally equal or exceed arrests and removals. There is a spike in 2012, most likely due to the CDS reclassification rather than an increase in border activity. After 2012, all three metrics steadily declined through the Obama years, with a secondary increase around 2018-2019 before the sharp drop in 2020 due to COVID-19 (Chishti et al., 2020) .

This chart directly answers our third question about whether the system became more enforcement-focused over time. The Trump administration (2017-2021), which campaigned on aggressive immigration enforcement, produced lower ICE removal counts than the Obama administration between 2012-2016 (Chishti et al., 2017). The data does not support the conclusion that the system became more enforcement-focused in terms of outcomes.

Principle Component Analysis of Immigration Features

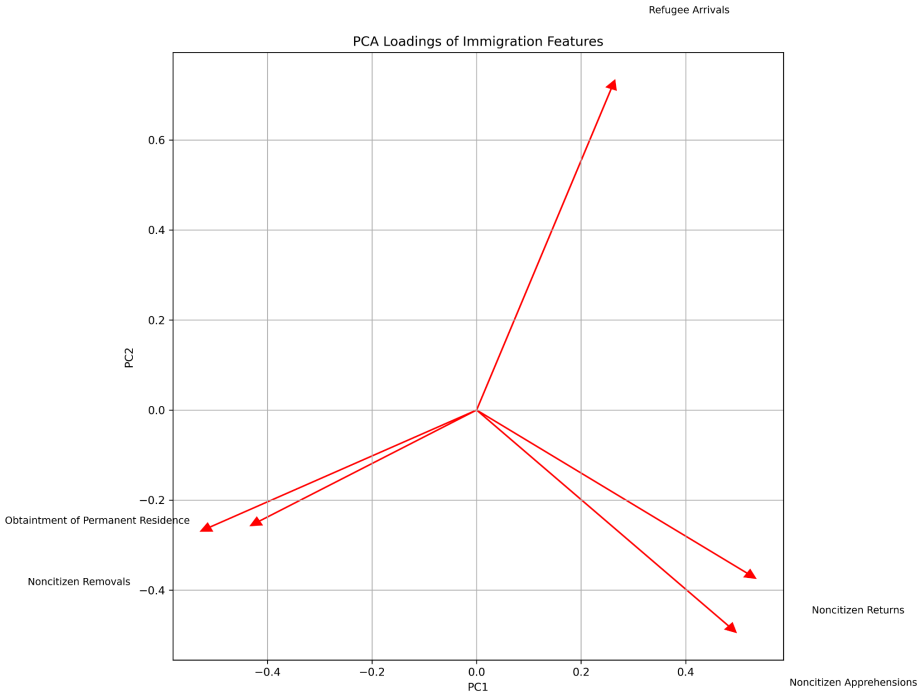

Figure 6. PCA Loadings of Immigration Features: Full Period (1980–2021). Apprehensions, removals, and returns load strongly onto PC1, while refugee arrivals load almost exclusively onto PC2, indicating two structurally independent dimensions.

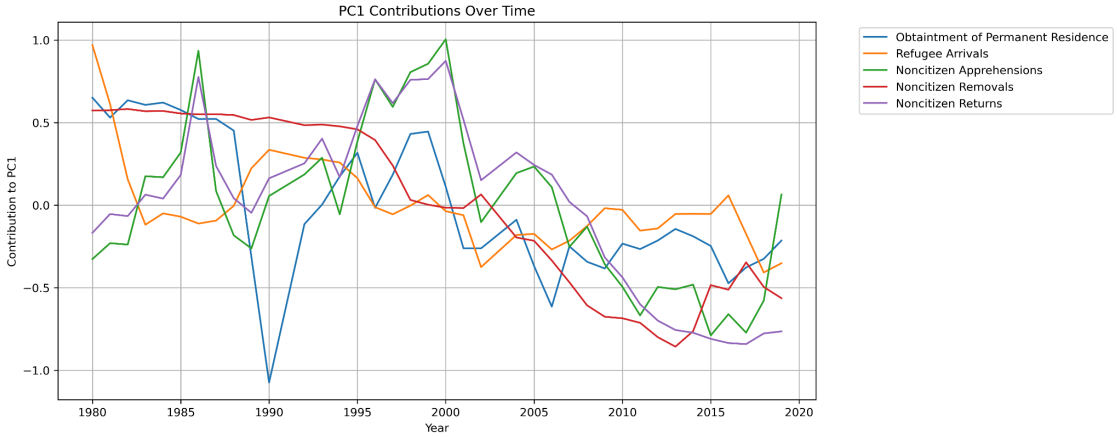

Figure 7. Immigration Features Contribution to PC1 Over Time (1980–2021). Contributions fluctuate across decades with notable volatility around 1990, reflecting shifting enforcement priorities across administrations.

PCA applied to the full 1980-2021 DHS dataset explains about 81% of the total variance in the five immigration features across just two principal components. indicating that most of the complexity in four decades of U.S. immigration data tells us two things. One is that apprehensions and returns move together and define PC1. The other is that refugee arrivals operate almost entirely independently on PC2, uncorrelated with the other variables. Obtainment of Permanent Residency and noncitizen removals also loads into PC1 but in opposing directions than that of noncitizen returns and apprehensions. This fundamentally answers our first question.

The single biggest factor shaping immigration patterns is enforcement intensity while refugee arrivals are driven by something else entirely (global events, international obligations, or humanitarian needs).

Permanent residency obtainment also loads onto PC1 but in the opposite direction than enforcement. This means enforcement-heavy years are systematically associated with fewer people successfully obtaining lawful status. This pattern can be directly tied to our second question about whether fewer people immigrate when deportation rates are high. According to the legal admissions levels, the answer appears to be yes.

Era-Specific PCA: Obama (2009-2017)

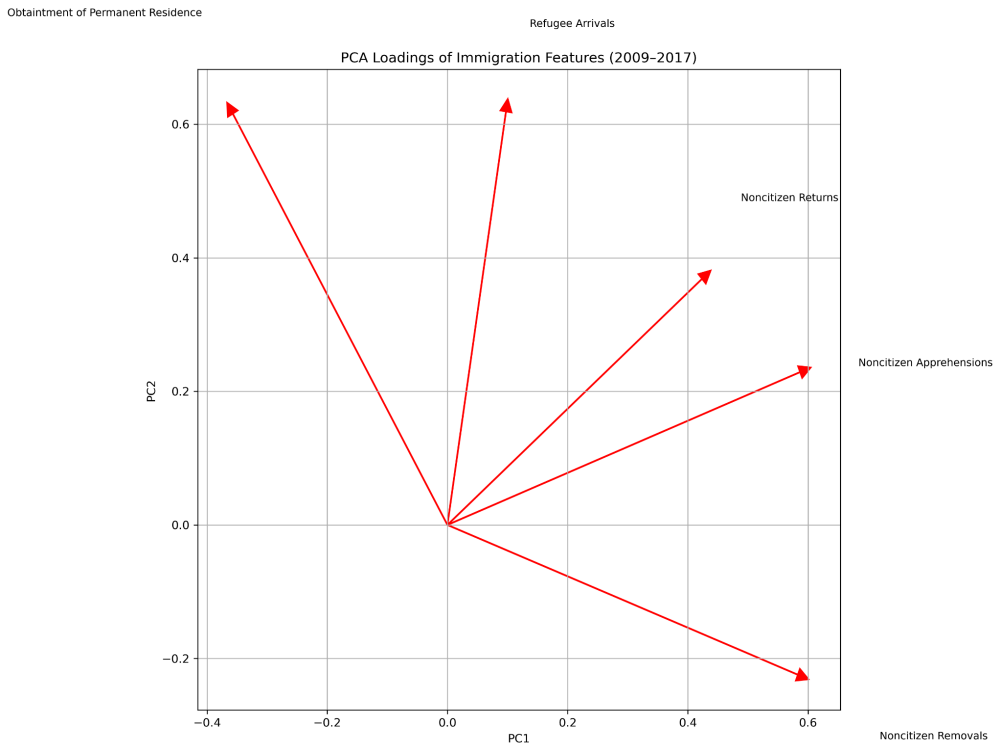

Figure 8. PCA Loadings of Immigration Features: Obama Era (2009–2017). Permanent residency and refugee arrivals load onto PC2, while enforcement variables cluster along PC1, reflecting a period of high but declining enforcement intensity.

When PCA is restricted to 2009-2017, the loading structure shows that noncitizen apprehensions and returns are positively correlated, both pointing in the same PC1 direction. Noncitizen removals load separately, while permanent residency obtainment and refugee arrivals sit in distinct positions reflecting their relative independence from enforcement during this period.

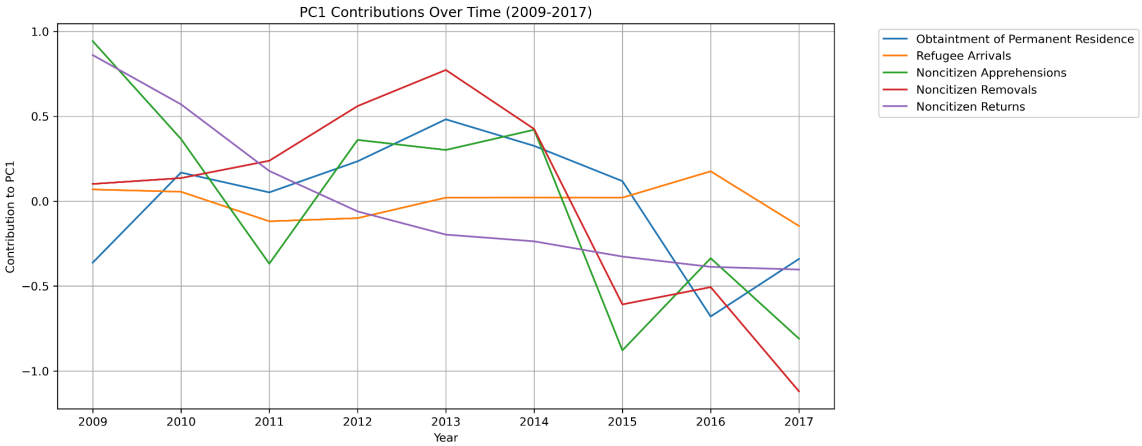

Figure 9. Immigration Features Contribution to PC1 Over Time (2009–2017). Contributions are relatively stable early in the period before diverging after 2014, consistent with the implementation of DACA.

The PC1 contribution plot tells a year-by-year version of the enforcement story. No single feature consistently dominates, which reflects the mixed enforcement signals of the Obama years when the administration was simultaneously setting deportation records and implementing DACA. The most notable event is the sharp spike in noncitizen removals in 2013, directly coinciding with the full expansion of Secure Communities. After 2014, apprehensions and removals declined in step with DACA, as enforcement priorities narrowed away from broad removal toward individuals with serious criminal records.

Era-Specific PCA: Trump (2017-2021)

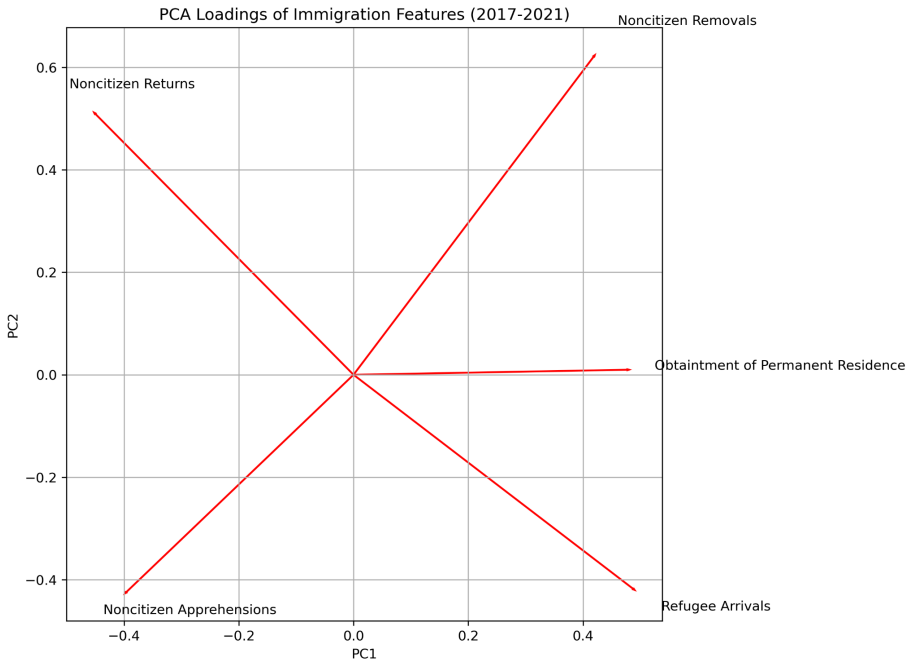

Figure 10. PCA Loadings of Immigration Features: Trump Era (2017–2021). Features split into two opposing pairs along PC1, with removals and refugee arrivals moving in opposite directions, reflecting a period of shifting enforcement and admission dynamics.

The 2017-2021 loading structure looks different from the Obama-era PCA plots. Noncitizen removals and returns from a cross pattern with noncitizen apprehensions, suggesting these three enforcement metrics moved in more independent directions rather than uniformly. Permanent residency obtainment is nearly flat along PC2, while refugee arrivals point strongly downward, consistent with the sharp refugee admission cuts made by the Trump administration.

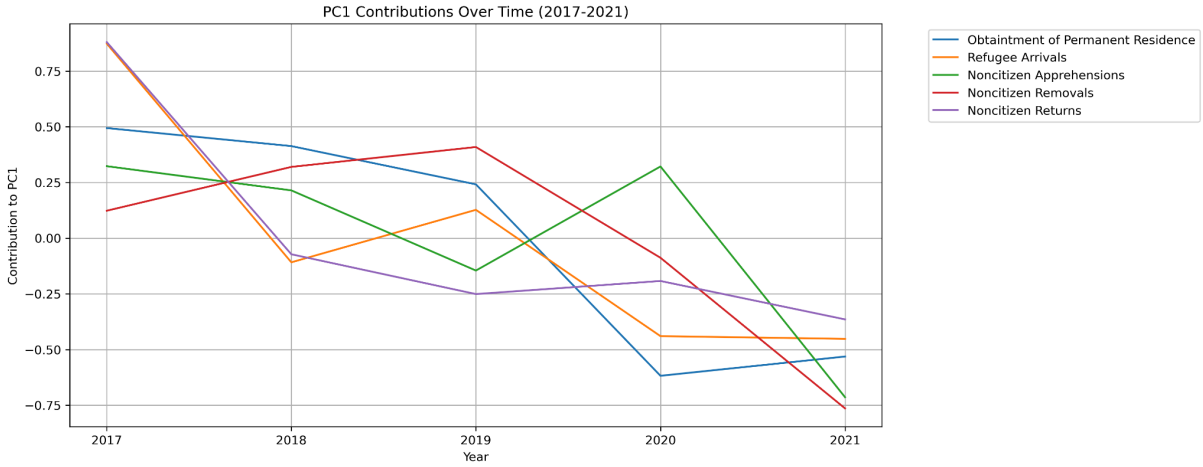

Figure 11. Immigration Features Contribution to PC1 Over Time (2017–2021). All features decline steadily from 2017, with a sharp drop in 2020 consistent with COVID-19 disruptions across all enforcement and admission channels.

The PC1 contribution plot shows all features starting positively in 2017, reflecting an initial broad enforcement posture across all immigration metrics simultaneously. They then generally decline over the period, with a sharp uniform drop in 2020. What the PCA makes visible here that the raw numbers do not is the coherence of the 2017 starting point as everything moved together at the start of its administration, suggesting a system-wide enforcement signal rather than targeted changes to individual pathways.

Principal Component Regression (PCR)

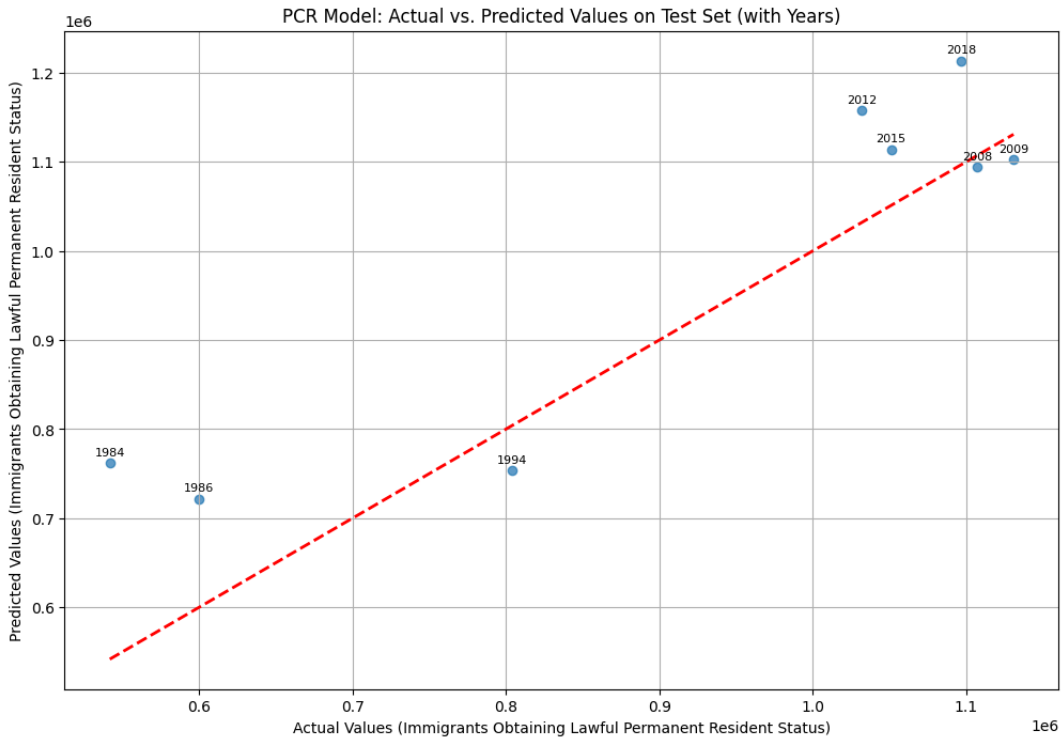

Figure 12. PCR Model on Years 1980 - 2021: Actual vs. Predicted Values on Test Set. The dashed line represents a perfect prediction. Points above the line are overpredicted; points below are underpredicted.

We extended the PCA into a Principal Component Regression model to directly quantify the relationship between enforcement intensity and immigrants obtaining lawful permanent resident status - putting a number on what the PCA loadings implied qualitatively.

$$\mathrm{PRS} = 917,686.3015 - 75,889.6575 \cdot \mathrm{PC}_1 - 44,673.7883 \cdot \mathrm{PC}_2$$

Regression line

Both components carry negative coefficients, meaning that higher scores on either dimension - more enforcement activity or more refugee arrivals - are associated with fewer permanent residency grants.

The two components explain 91% of the variance in the input features, and the PCR model has an R^2 value of 0.61 on the test set. While the test set is small at 7 years, the coefficient on PC1 implies that years with noticeably higher enforcement activity tend to have roughly 75,000 fewer lawful permanent residency grants. In a year where approximately one million people obtain permanent residency, that is a 7.5% reduction that can be attributed to enforcement intensity alone - and this answers our second question as this means that higher enforcement is associated with a considerable amount of fewer people obtaining legal permanent residency, possibly making migrants double guess themselves on the thought of them moving to the US. The PC2 coefficient also confirms that refugee arrivals contribute to permanent residency through an entirely different channel, one that enforcement has no power over.

GDP per Capita and Removal Patterns

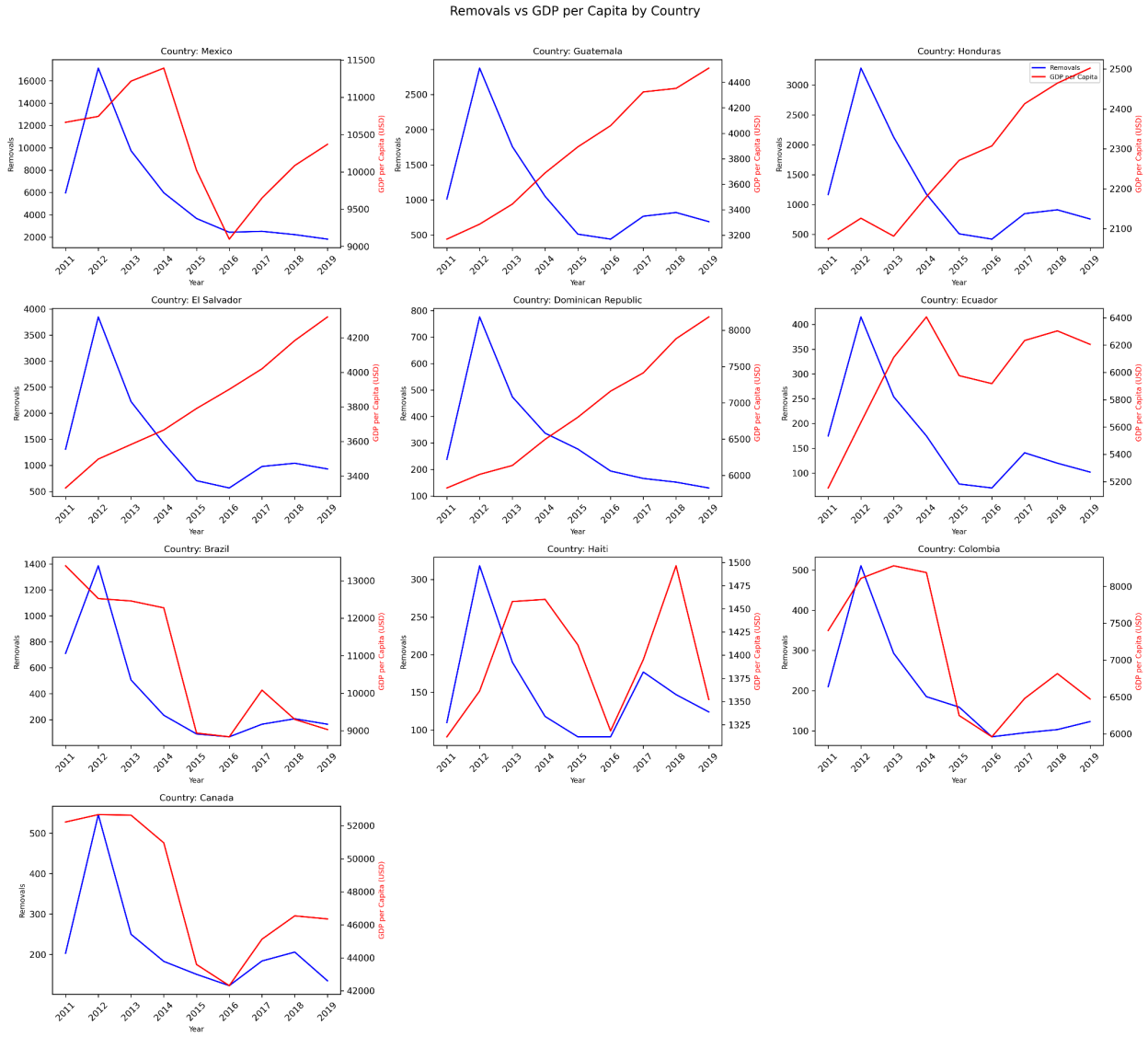

Figure 13. Removals vs. GDP per Capita by Country (2011–2019, top 10 countries by removal count). Removals are in blue generally while GDP per capita is in red.

To address our first question to a deeper extent and to figure out what factors affect immigration the most, we looked beyond domestic policy and asked whether economic conditions in origin countries predict removal volumes. If economic hardship is the primary driver of migration, we would expect countries with lower or declining GDP per capita to show higher removal counts, since economic pressure pushes more people to migrate in the first place. We merged ICE removal counts by country and year with World Bank GDP per capita data, assuming departure country as a proxy for origin country.

The top ten countries by total removals across the dataset are Mexico, Guatemala, Honduras, El Salvador, Dominican Republic, Ecuador, Brazil, Haiti, Colombia, and Canada. All but one are in Latin America, with Mexico leading by a very large margin at over one million removals. We plotted removal counts against GDP per capita over time for each of these ten countries individually, excluding COVID-19 years (2020-2021) to avoid distortion.

Across all ten countries, there is no consistently strong correlation between GDP per capita and removal counts. Mexico shows a modest negative trend as GDP gradually rises, removals decline, but this relationship is confounded by the broader post-2012 enforcement decline visible across all countries. For Guatemala, Honduras, and El Salvador, removal volumes track more closely with U.S. enforcement policy cycles than with domestic GDP fluctuations.

The absence of a consistent GDP-removal relationship across countries suggests that removal volume is driven more by geographic proximity to the United States than by year-to-year changes in economic conditions in origin countries. While economic push factors are theoretically important motivators of migration, our data captures only the enforcement outcomes rather than the migration decisions themselves, which limits what we can infer about economic motivations.

Visa Issuance Analysis

For our third & last research questions, whether the system has become more enforcement heavy or admission focused, we decided to look at the admission side directly with Visa statistics from the US Department of State as enforcement data alone cannot answer this question. The dataset was cleaned by aggregating all issuing offices within each country into a single annual value per country. We then merged it with ICE removal counts for the top ten countries by removals, plotting both variables over time for each country to look for a possible enforcement-admission tradeoff.

Figure 14. Yearly Visa Issuances vs. Yearly Removals by Country (2015–2021, only top 10 countries with the most removals)

As seen in Figure 14, for the majority of countries, visa issuances outnumber removals by so much that removals are nearly invisible relative to the visa issuances for countries like Brazil, Colombia, Ecuador, and Mexico. Guatemala and Honduras are the only exceptions where removal counts are similar to the visa issuance levels, especially around 2019 - 2020, the heightened enforcement activity towards Central American countries during that time possibly playing a big role. Rather than showing an enforcement-admission tradeoff, the data suggests that both channels were shaped by the same external forces. Meaning that the system from 2015

- 2021 did not seem to appear to have made a structural shift away from admission towards enforcement. If anything, the scale difference from this visualization suggests that legal admission activity far exceeds enforcement activity, answering our third research question.

Conclusion

Through our analysis of data from ICE, Deportation Data Project, Department of Homeland Security, the State Department, and the World Bank API, we sought to understand what enforcement patterns correlate with, whether that be policy events, economic conditions, or admission-side trends.

On the enforcement side, the evidence is pretty clear. The K-Means clustering identified three distinct enforcement time periods that aligns directly with the Secure Communities Program, DACA, and COVID-19. The PCA confirms that enforcement intensity is the biggest factor shaping immigration patterns, with apprehensions, removals, and returns moving together in one dimension. The PCR model quantified this further by showing that enforcement-heavy years are associated with roughly 75,000 fewer lawful permanent residency grants, a 7.5% reduction that directly ties into our second research question. Refugee arrivals, however, is an entirely independent dimension throughout all eras, showing that refugees are generally a humanitarian and global factor rather than a domestic enforcement one.

On the economic side, the GDP per capita analysis did not reveal a consistent relationship between economic conditions in departure countries and removal rates. Removal patterns appear to be impacted heavier by geographic proximity and US policy cycles than by annual economic fluctuations in origin countries, a meaningful result that suggests migration is a far more complex issue than how much origin countries are making.

For our third research question, the visa issuance analysis showed that legal admission activity far exceeded that of enforcement activity at the country level for nearly every top 10 countries with the most removals. The system didn’t appear to have a clear structural shift away from admission toward enforcement during the 2015 - 2021 period.

A key limitation of our analysis is that we relied on removal data as a comparable indicator for immigration activity since data on individuals voluntarily choosing to immigrate is not available. The ICE dataset also covers a limited time range of 2011 to 2021, even though they formed in 2003, we didn’t have access to that data, limiting the patterns we can observe. For future iterations, we would like to develop a predictive model incorporating a wider range of features, including push factors from origin countries such as crime rates and political instability, to capture the side of the story our current data cannot tell.

Citations

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